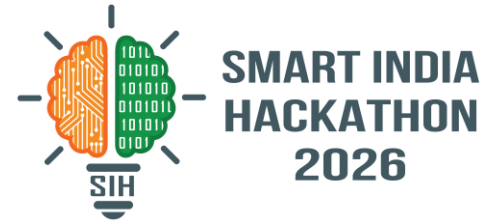


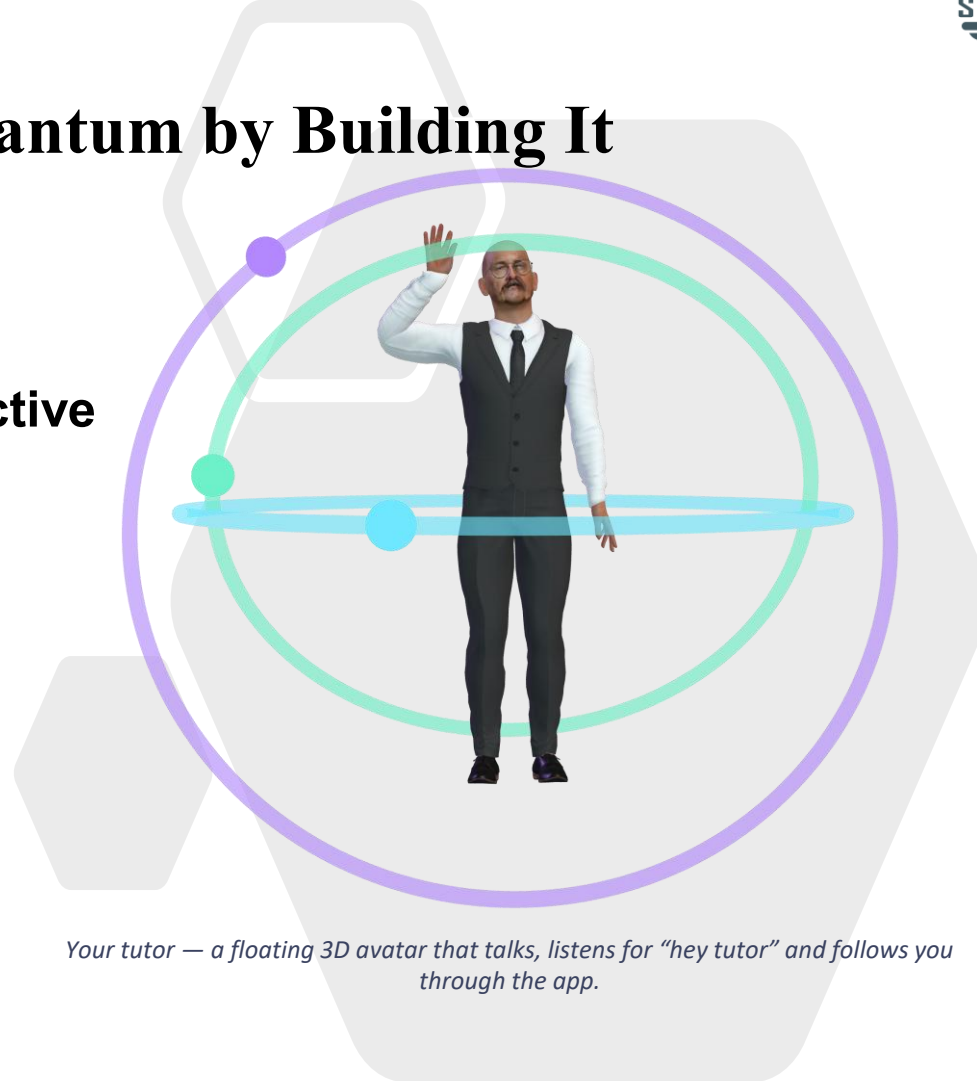


SMART INDIA HACKATHON 2026



Qurios — Learn Quantum by Building It

- Problem Statement ID – SIH26140
- Problem Statement Title – AI-Based Interactive Quantum Algorithm Learning Platform
- Theme – Smart Education
- PS Category – Software
- Team ID –
- Team Name – X FORCE



Your tutor — a floating 3D avatar that talks, listens for “hey tutor” and follows you through the app.

IDEA TITLE

Qurious — a quantum lab, a self-checking AI tutor and a real-machine noise simulator in one app. Runs in a browser, and offline as a 7.4 MB Android APK.

01 THE PROBLEM

- Quantum ideas are abstract: slides show the algebra, never the state moving.
- Existing tools are either rigorous but cloud-locked, or instant but teach nothing.
- Real hardware means an account, a queue and English-only material.
- An AI helper that invents a plausible wrong answer teaches the wrong physics.

02 THE SOLUTION

- Drag gates onto a canvas; a full statevector simulator runs on the device.
- Editor in OpenQASM 3, Qiskit, Cirq and PennyLane — canvas and code are one circuit.
- A tutor that explains, debugs and repairs circuits, speaks aloud, and works offline.
- 8 lessons, 15 challenges, 19 quiz questions, mastery tracking and a teacher view.

03 INNOVATION

- Grounded AI: every claim is run on the simulator first and discarded if it disagrees.
- Grading on physics, not string matching — fidelity that ignores global phase.
- One circuit model drives canvas, code, four SDKs, grader, tutor and noise.
- Fully offline: 105-topic tutor knowledge base, 12 qubits on a mid-range phone.

THE ONE THING NOTHING ELSE TEACHES — THE NOISE LAB

- The same circuit, run the way a real machine runs it: Pauli errors, amplitude damping, dephasing and readout flips, as Monte Carlo trajectories.
- Bell state at today's superconducting error rates: $|01\rangle$ and $|10\rangle$ — **outcomes ideal physics forbids — appeared 63 times in 1,024 shots.**
- Four presets: Ideal, Trapped ion, Today's superconducting, Early hardware.



Live output from the working build — not a mock-up.

TECHNOLOGIES TO BE USED

Frontend — React 18 + TypeScript, Vite, Three.js (WebGL), CodeMirror 6

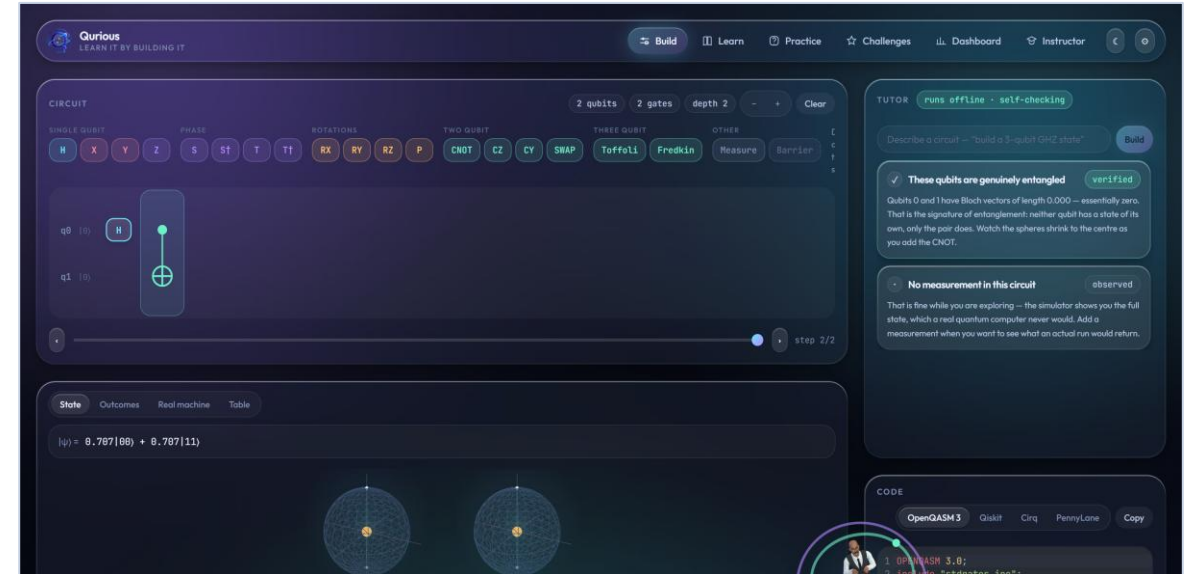
Quantum core — Own statevector engine, 12 qubits on-device; Qiskit Aer, PennyLane, Cirq for export and cross-checking

AI tutor — Pluggable LLM (Groq, Gemini, OpenRouter, Ollama) over a 105-topic offline base; Bayesian Knowledge Tracing

Speech & 3D — Web Speech API, “hey tutor” wake word, rigged GLB avatar with nine animation clips

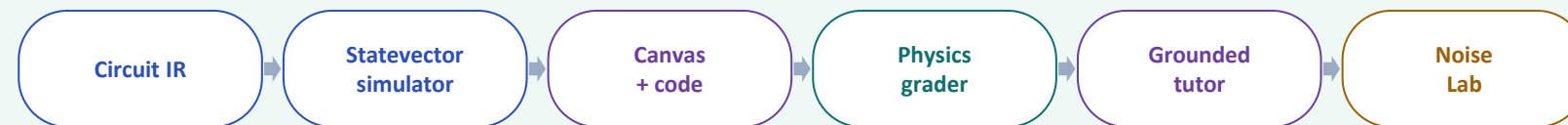
Mobile — Capacitor → Android APK (minSdk 22, 7.43 MB) and an offline-first PWA

Server (optional) — FastAPI + Qiskit Aer for circuits beyond the on-device limit



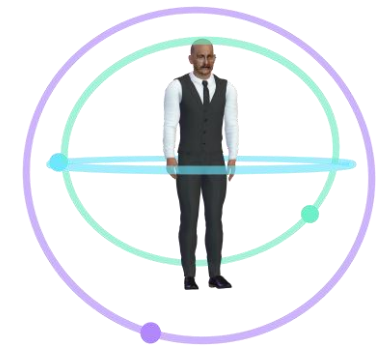
The build today: canvas, live state, four-SDK code view and the self-checking tutor.

METHODOLOGY — ONE CIRCUIT MODEL, EVERY VIEW



The IR is the single source of truth. Edit the canvas or edit the code — both write the same object, so the two can never drift apart, and the grader, the tutor and the noise model all read it.

Verification. 22 circuits are cross-checked against Qiskit Aer; the worst disagreement is 6.66×10^{-16} . 146 automated tests pass.



Installable APK — with the tutor on top.

TECHNICAL FEASIBILITY

- It already runs: web app plus an installable 7.43 MB APK, not a proposal.
- 146 automated tests; the simulator matches Qiskit Aer to 6.66×10^{-16} .
- 12 qubits simulate in a browser tab — no GPU, no server, no account.
- Noise uses Monte Carlo trajectories, so cost stays 2^n and never 4^n .

ECONOMIC FEASIBILITY

- Entirely open-source stack — React, Three.js, Qiskit, PennyLane, Cirq. No licences.
- Compute happens on the learner's own device: zero marginal cost per student.
- The LLM is optional and bring-your-own-key; free tiers cover a whole class.
- Distribution is a URL or a 7.4 MB file. No lab, no hardware, no cloud bill.

EDUCATIONAL VIABILITY

- Built for Tier-2 and Tier-3 colleges that will never have a quantum lab.
- Lessons and challenges are data files, so a teacher can author content.
- The instructor view reports per-concept mastery across a class.
- Serves students, teachers and working engineers reskilling into quantum roles.

POTENTIAL CHALLENGES AND THE STRATEGY FOR EACH

Risk. On-device simulation stops at about 12 qubits.

Strategy. Every lesson and challenge is designed below that ceiling, the limit is shown in the interface rather than hidden, and an optional FastAPI + Aer service takes larger circuits.

Risk. A language model can state something false with confidence.

Strategy. Nothing reaches the learner unchecked: the tutor proposes, the simulator executes, and the claim is dropped when the output disagrees. With no API key it falls back to 105 written topics.

Risk. We do not run on real quantum hardware.

Strategy. The Noise Lab teaches what hardware actually does to a circuit — arguably more than one queued job would. A vendor runtime adapter is an addition later, never a dependency.

Risk. Low-end phones, weak networks, mixed languages.

Strategy. Offline-first PWA with a WebGL budget and 2D fallbacks, tested at 412×892 ; speech and tutor content available in Indian languages.

WHO IT REACHES

- Students in colleges with no quantum laboratory — the ordinary case in Tier-2 and Tier-3 India.
- Teachers who need per-concept mastery data without first building a course.
- Self-learners on a phone: offline, in their own language, with no account and no queue.
- Working engineers reskilling into quantum roles alongside a job.

WHAT ACTUALLY CHANGES FOR THEM

- Concepts stop being algebra: superposition, phase and entanglement become things you move.
- Feedback in seconds — physics-based grading, and a tutor that repairs the circuit and says why.
- Learners meet noise on day one instead of after their first disappointing hardware run.
- The work transfers: every circuit exports to Qiskit, Cirq, PennyLane and OpenQASM 3.

EDUCATIONAL

- Active learning instead of watching.
- Instant, physics-based feedback.
- Sequencing personalised by knowledge tracing.

SOCIAL

- Works offline, on a mid-range phone.
- Tutor speaks Indian languages.
- No account, no queue, no gatekeeping.

ECONOMIC

- No laboratory and no hardware spend.
- No licences and no per-student cloud bill.
- A 7.4 MB file is the whole deployment.

STRATEGIC

- Builds quantum literacy at college scale.
- Supports the National Quantum Mission's skilling aims.
- Graduates arrive already fluent in Qiskit.

Measured today, in the build you can install: 8 lessons · 15 challenges · 19 quiz questions · 105 offline tutor topics · 4 SDK exports · 4 noise presets · 146 tests green

DETAILS AND LINKS OF THE REFERENCE AND RESEARCH WORK

[1] Systematic Review of Quantum Machine Learning in Education 5.0

Algorithms, MDPI (2026) — doi.org/10.3390/a19050379 — Establishes the demand for practical, interactive quantum education.

[2] Teaching Quantum Computing within K–12: a systematic review

University of Eastern Finland repository (2023) — Evidence that visual, hands-on tools outperform lecture-only delivery.

[3] Quantum Computing Fundamentals — IBM Quantum Learning

learning.quantum.ibm.com — Reference treatment of qubits, superposition, entanglement and interference.

[4] Bits, Gates and Circuits — IBM Quantum Learning

learning.quantum.ibm.com — Source for gate conventions and circuit construction.

[5] Corbett & Anderson, Knowledge Tracing

User Modeling and User-Adapted Interaction, 4(4), 1994 — The Bayesian model behind our per-concept mastery estimates.

[6] Nielsen & Chuang, Quantum Computation and Quantum Information

Cambridge University Press — Gate definitions, the Bloch sphere and the conventions our engine follows.

[7] Qiskit Aer — IBM

qiskit.github.io/qiskit-aer — Used as the independent reference simulator for our cross-check.

[8] Competitive survey of existing quantum-learning tools

Team research, September 2026 — 13 products across 5 clusters — The basis for the capability comparison on the right.

WHAT WE BENCHMARKED AGAINST

- IBM Quantum Learning + Composer
- Qiskit Studio (AI4quantum) — closest analogue
- Quirk · Quantum Odyssey
- PennyLane Codebook · Brilliant

Curious answers 20 of 21 capabilities. The one it does not is direct access to real quantum hardware — a deliberate trade, argued in the comparison report.

HOW TO CHECK OUR CLAIMS

146 automated tests

```
npx vitest run
```

Simulator vs Qiskit Aer

```
python server/crosscheck.py
```

Worst disagreement across 22 circuits: 6.66×10^{-16}